DevOps Lab - 1

Task 1: Record All Commands

Objective

To record and document all commands, pipeline configurations, screenshots, commit history, and observations performed during the DevOps experiments.

Commands Used

The following commands were used during the implementation:

Java verification

```
admin1@DESKTOP-BBG39NK:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
```

Git installation and verification

```
admin1@DESKTOP-BBG39NK:~$ sudo apt install git -y
git is already the newest version (1:2.53.0-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 159
admin1@DESKTOP-BBG39NK:~$ git version
git version 2.53.0
```

Git configuration

```
admin1@DESKTOP-BBG39NK:~$ git config --global user.name "gurleen-51"
admin1@DESKTOP-BBG39NK:~$ git config --global user.email "silvydadiala@gmail.com"
admin1@DESKTOP-BBG39NK:~$ git config --global --list
user.name=gurleen-51
user.email=silvydadiala@gmail.com
```

Create project directory

```
admin1@DESKTOP-BBG39NK:~$ mkdir devops-ci-cd-lab
admin1@DESKTOP-BBG39NK:~$ cd devops-ci-cd-lab
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano app.py
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 app.py
Hello, Gurleen!
```

Git repository initialization and Check repository status

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/admin1/devops-ci-cd-lab/.git/
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py

nothing added to commit but untracked files present (use "git add" to track)
```

Add files

Commit changes

Create and switch to development branch

Switch to main branch

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py

nothing added to commit but untracked files present (use "git add" to track)
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git add .
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m " DevOps application"
[master (root-commit) 7be3151]  DevOps application
 1 file changed, 1 insertion(+)
 create mode 100644 app.py
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git log --oneline
7be3151 (HEAD -> master)  DevOps application
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git branch development
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git switch development
Switched to branch 'development'
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git add .
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git commit -m "Update application greeting"
On branch development
nothing to commit, working tree clean
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git switch main
```

Merge development branch

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git merge development
Already up to date.
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git log --oneline --graph --all
* 7be3151 (HEAD -> development, master)  DevOps application
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git remote add origin https://github.com/gurleen-51/devops-ci-cd-lab.git
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git remote -v
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (fetch)
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (push)
```

Push changes to GitHub

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git remote -v
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (fetch)
origin  https://github.com/gurleen-51/devops-ci-cd-lab.git (push)
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git push -u origin main
Username for 'https://github.com': gurleen-51
Password for 'https://gurleen-51@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 243 bytes | 243.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gurleen-51/devops-ci-cd-lab.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ cd ~/devops-ci-cd-lab
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ ls
app.py
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Run Python test

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano test_app.py
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ cat test_app.py
def test_application():
    assert True
```

Run deployment script

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ sudo apt install python3-pytest -y
Installing:
  python3-pytest

Installing dependencies:
  python3-iniconfig  python3-pluggy

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 160
  Download size: 305 kB
  Space needed: 1545 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-iniconfig all 2.1.0-2 [6962 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pluggy all 1.6.0-2 [21.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pytest all 9.0.2-4 [277 kB]
Fetched 305 kB in 5s (57.4 kB/s)
Selecting previously unselected package python3-iniconfig.
(Reading database ... 38075 files and directories currently installed.)
Preparing to unpack .../python3-iniconfig_2.1.0-2_all.deb ...
Unpacking python3-iniconfig (2.1.0-2) ...
Selecting previously unselected package python3-pluggy.
Preparing to unpack .../python3-pluggy_1.6.0-2_all.deb ...
Unpacking python3-pluggy (1.6.0-2) ...
Selecting previously unselected package python3-pytest.
Preparing to unpack .../python3-pytest_9.0.2-4_all.deb ...
Unpacking python3-pytest (9.0.2-4) ...
Setting up python3-iniconfig (2.1.0-2) ...
Setting up python3-pluggy (1.6.0-2) ...
Setting up python3-pytest (9.0.2-4) ...
Processing triggers for man-db (2.13.1-1build1) ...
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest --version
pytest 9.0.2
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest
============================================================================= test session starts ==============================================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
```

```
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
collected 1 item

test_app.py .

============================== 1 passed in 0.01s ===============================
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ mkdir -p deployment
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ chmod +x deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ ./deployment/deploy.sh
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
```

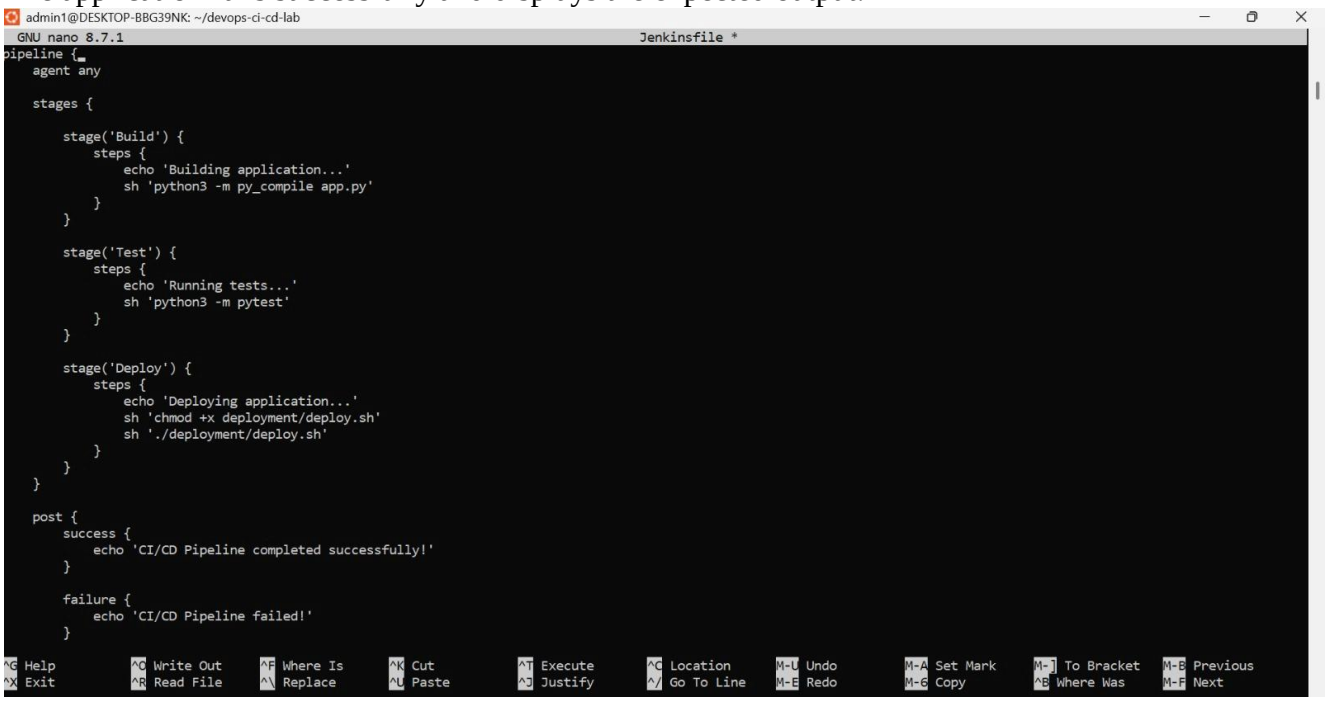
Jenkins Pipeline

The Jenkins pipeline was configured using a Jenkinsfile containing three major stages:

1. **Build**
2. **Test**
3. **Deploy**

The pipeline automatically retrieves the source code from GitHub and executes these stages.

The application runs successfully and displays the expected output.



```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ sudo apt install python3-pytest -y
Installing:
  python3-pytest

Installing dependencies:
  python3-iniconfig  python3-pluggy

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 160
  Download size: 305 kB
  Space needed: 1545 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-iniconfig all 2.1.0-2 [6962 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pluggy all 1.6.0-2 [21.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pytest all 9.0.2-4 [277 kB]
Fetched 305 kB in 5s (57.4 kB/s)
Selecting previously unselected package python3-iniconfig.
(Reading database ... 38075 files and directories currently installed.)
Preparing to unpack .../python3-iniconfig_2.1.0-2_all.deb ...
Unpacking python3-iniconfig (2.1.0-2) ...
Selecting previously unselected package python3-pluggy.
Preparing to unpack .../python3-pluggy_1.6.0-2_all.deb ...
Unpacking python3-pluggy (1.6.0-2) ...
Selecting previously unselected package python3-pytest.
Preparing to unpack .../python3-pytest_9.0.2-4_all.deb ...
Unpacking python3-pytest (9.0.2-4) ...
Setting up python3-iniconfig (2.1.0-2) ...
Setting up python3-pluggy (1.6.0-2) ...
Setting up python3-pytest (9.0.2-4) ...
Processing triggers for man-db (2.13.1-1build1) ...
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest --version
pytest 9.0.2
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
```

```
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
collected 1 item

test_app.py .

============================== 1 passed in 0.01s ===============================
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ mkdir -p deployment
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ chmod +x deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ ./deployment/deploy.sh
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
```

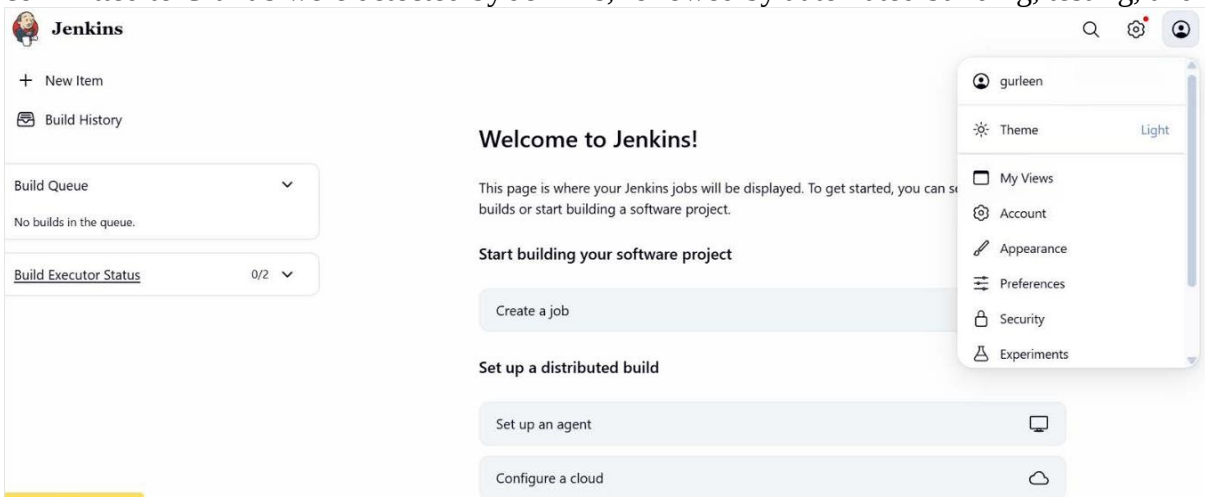
Automation Result

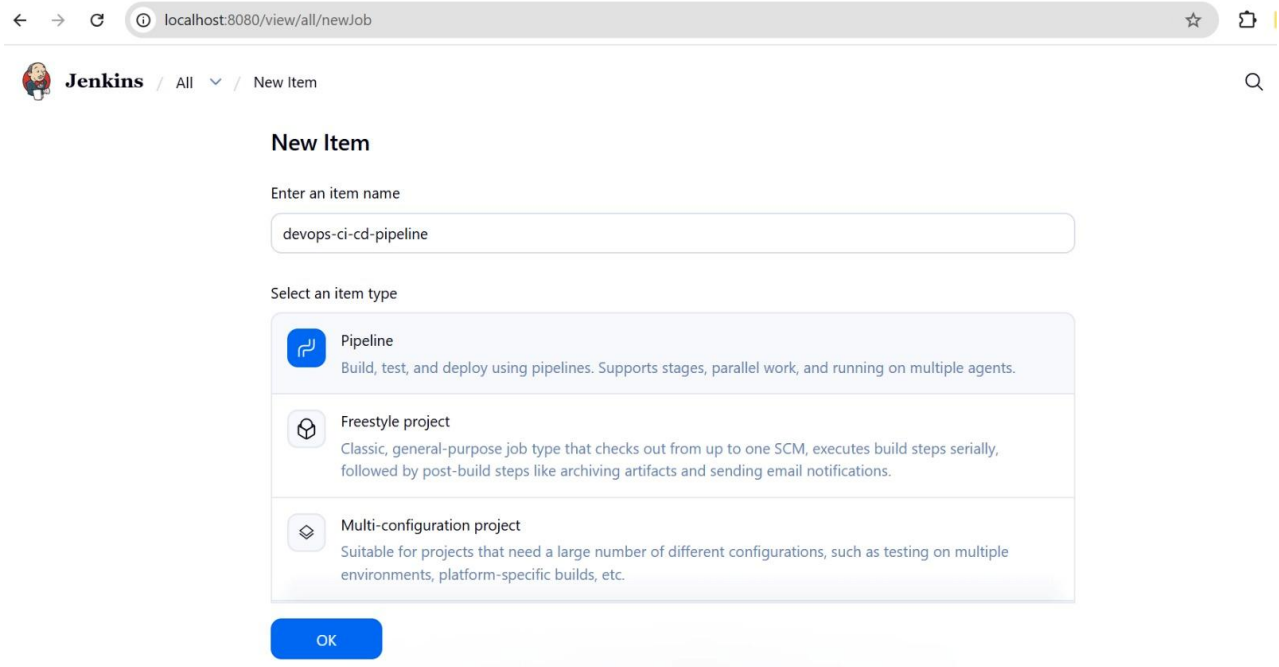
The pipeline was successfully automated. A change pushed to GitHub triggered a new Jenkins build through Poll SCM.

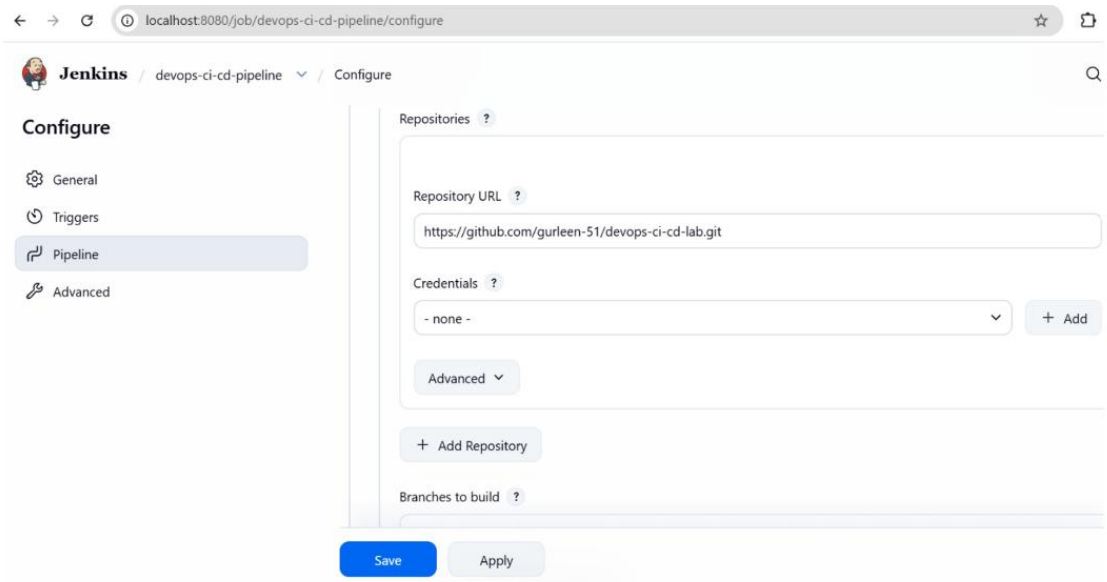
Jenkins automatically performed:

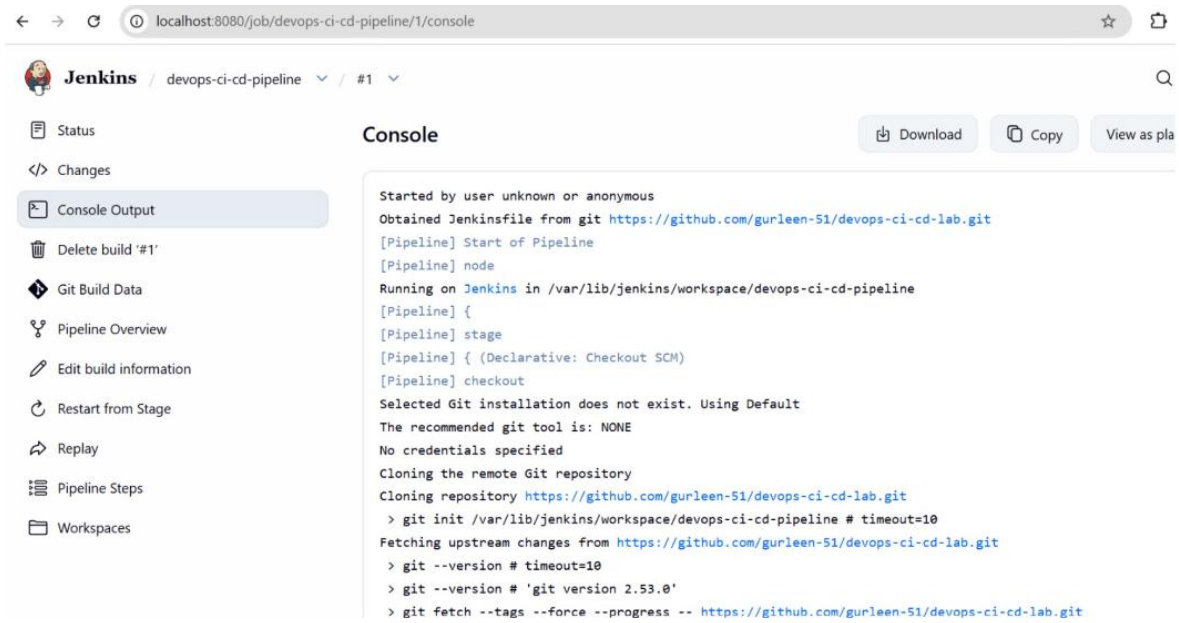
GITHUB CHANGE
↓
JENKINS TRIGGER
↓
BUILD
↓
TEST
↓
DEPLOY
↓
SUCCESS

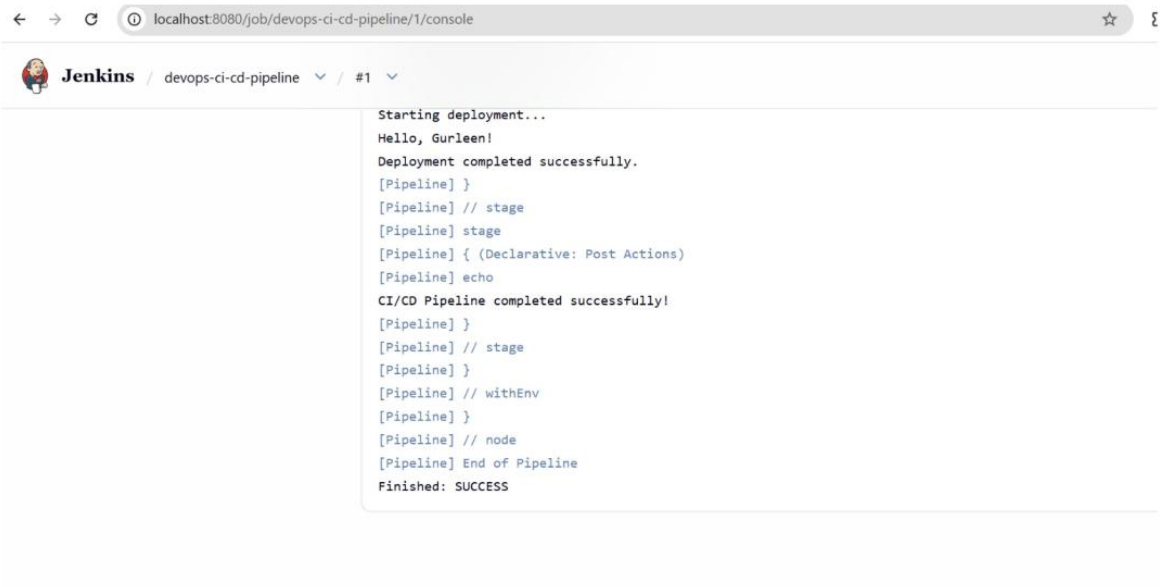
The CI/CD workflow successfully automated the software delivery process. Changes committed to GitHub were detected by Jenkins, followed by automated building, testing, and

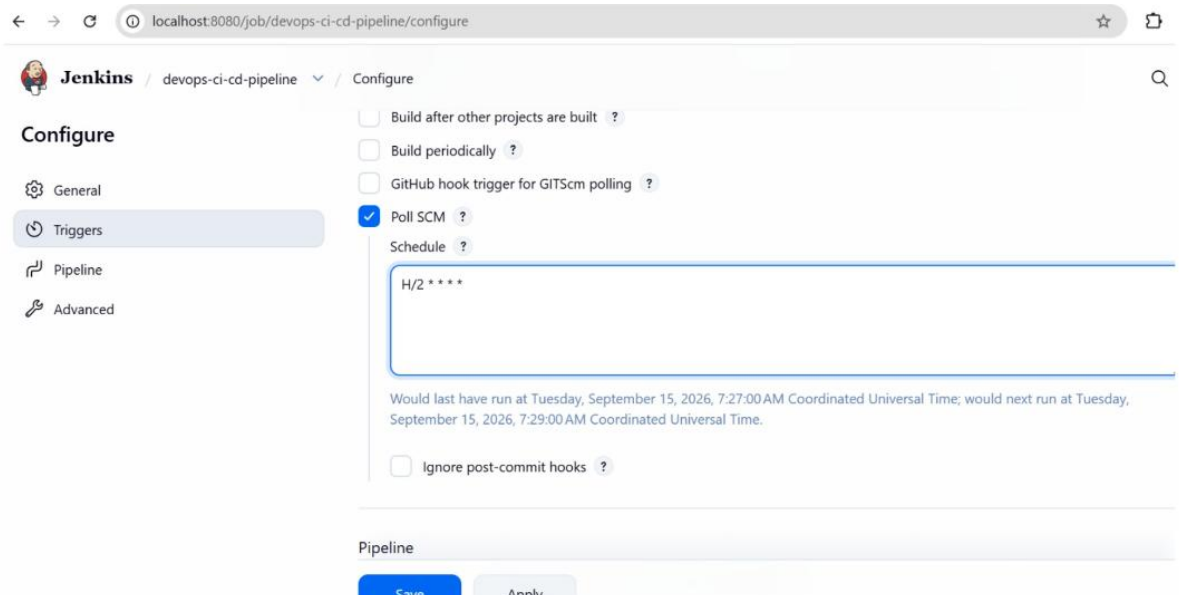


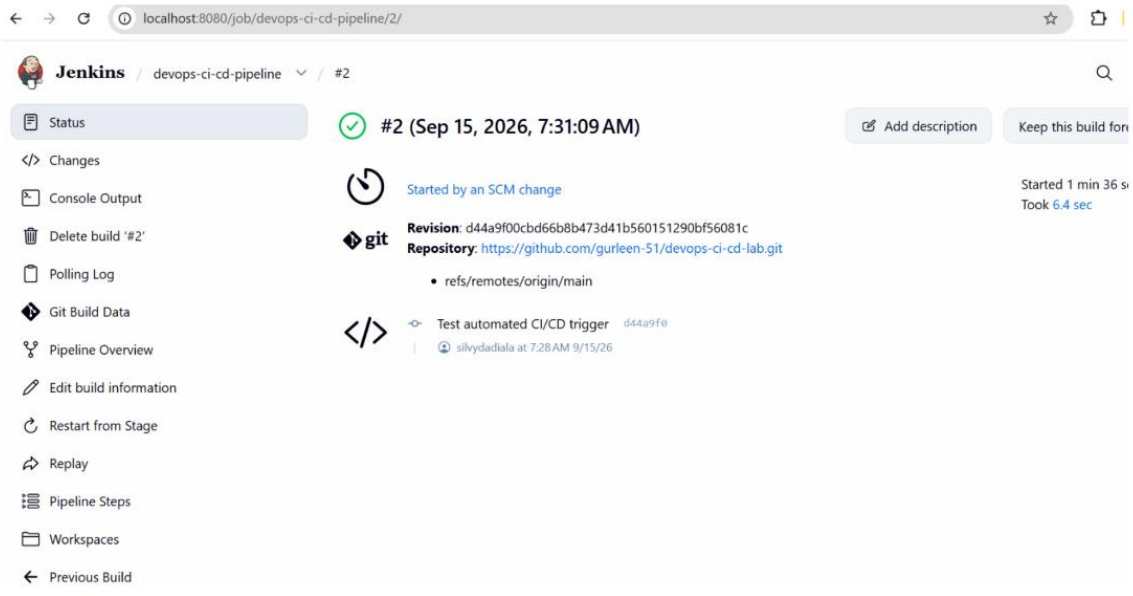



localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins / devops-ci-cd-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Repositories
Repository URL
https://github.com/gurleen-51/devops-ci-cd-lab.git
Credentials
- none -
+ Add
Advanced
+ Add Repository
Branches to build
Save
Apply


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Status
Changes
Console Output
Delete build '#1'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as pla
Started by user unknown or anonymous
Obtained Jenkinsfile from git https://github.com/gurleen-51/devops-ci-cd-lab.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-ci-cd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gurleen-51/devops-ci-cd-lab.git
> git init /var/lib/jenkins/workspace/devops-ci-cd-pipeline # timeout=10
Fetching upstream changes from https://github.com/gurleen-51/devops-ci-cd-lab.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/gurleen-51/devops-ci-cd-lab.git


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins / devops-ci-cd-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Schedule
H/2 * * * *
Would last have run at Tuesday, September 15, 2026, 7:27:00 AM Coordinated Universal Time; would next run at Tuesday, September 15, 2026, 7:29:00 AM Coordinated Universal Time.
Ignore post-commit hooks
Pipeline


localhost:8080/job/devops-ci-cd-pipeline/2/
Jenkins / devops-ci-cd-pipeline / #2
Status
Changes
Console Output
Delete build '#2'
Polling Log
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#2 (Sep 15, 2026, 7:31:09 AM)
Add description
Started by an SCM change
Started 1 min 36 s
Took 6.4 sec
Revision: d44a9f00cbd66b8b473d41b560151290bf56081c
Repository: https://github.com/gurleen-51/devops-ci-cd-lab.git
refs/remotes/origin/main
Test automated CI/CD trigger
silvydadiala at 7:28 AM 9/15/26

Install and Configure Git and Jenkins

